



UNIVERSITÉ DE KINSHASA
FACULTÉ DES SCIENCES ET TECHNOLOGIES



MENTION MATHÉMATIQUES, STATISTIQUE ET INFORMATIQUE

B.P. 190 KINSHASA XI

VOTRE TITRE SERA ICI

Par :

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Gradué en Sciences

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VOTEX

À ADAM et ÈVE

Remerciements

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VOTRE NOM

Introduction

Lorem ipsum, or lipsum as it is sometimes known, is dummy text used in laying out print, graphic or web designs. The passage is attributed to an unknown typesetter in the 15th century who is thought to have scrambled parts of Cicero's *De Finibus Bonorum et Malorum* for use in a type specimen book. It usually begins with : .

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Chapitre I

Généralités

Lorem ipsum, or lipsum as it is sometimes known, is dummy text used in laying out print, graphic or web designs. The passage is attributed to an unknown typesetter in the 15th century who is thought to have scrambled parts of Cicero's De Finibus [2], Bonorum et Malorum for use in a type specimen book. It usually begins with.

I.1 Rappels

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I.1.1 Définition (Problème Primal de PLSI) une généralisation de la notion de programme linéaire avec la possibilité que soit le nombre de variables est infini, soit le nombre de contraintes est infini.

Nous allons considérer le cas où le nombre des contraintes est infini. Dans ce cas, le problème primal de PLSI peut être formulé comme suit

$$(P) = \begin{cases} \min & c^T x \\ S/C & a_t^T x \geq b_t, t \in T \\ & x \geq 0 \end{cases} \quad (I.1)$$

où $c \in \mathbb{R}^n$, T est un ensemble infini arbitraire, $a_t \equiv a(t) = (a_1(t), \dots, a_n(t))$ est une application de T dans $\mathbb{R}^n$, $b_t = b(t)$ est une fonction réelle sur T .

Les notations $c^T x$ et $a_t^T x$ désignent le produit scalaire usuel de $\mathbb{R}^m$ et la notation a_t^T désigne la transposée du vecteur a_t de $\mathbb{R}^n$.

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I.2 Contexte

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I.2.1 Théorème *À supposer que tout sous-système fini de A admet une solution. Alors A n'admet aucune solution si et seulement si il existe $\sum \in \mathbb{R}^n$ tel que $\sum \cdot x \geq N$ peut être déduite de A pour tout N .*

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Chapitre II

Deuxième chapitre

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II.1 Rappels

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II.1.1 Définition (Problème Primal de PLSI) Un programme linéaire semi-infini est une généralisation de la notion de programme linéaire avec la possibilité que ou bien le nombre de variables est infini, ou bien le nombre de contraintes est infini.

Nous allons considérer le cas où le nombre des contraintes est infini. Dans ce cas, le problème primal de PLSI peut être formulé comme suit

$$(P) = \begin{cases} \min & c^T x \\ S/C & a_t^T x \geq b_t, t \in T \\ & x \geq 0 \end{cases} \quad (II.1)$$

où $c \in \mathbb{R}^n$, T est un ensemble infini arbitraire, $a_t \equiv a(t) = (a_1(t), \dots, a_n(t))$ est une application de T dans $\mathbb{R}^n$, $b_t = b(t)$ est une fonction réelle sur T .

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II.2.1 Théorème *À supposer que tout sous-système fini de A admet une solution. Alors A n'admet aucune solution si et seulement si il existe $\sum \in \mathbb{R}^n$ tel que $\sum \cdot x \geq N$ peut être déduite de A pour tout N .*

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Conclusion

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Bibliographie

- [1] ALEJANDRO WANES, SAM B. NADLER JR : ***Hyperspaces : Fundamentals and Recent Advances***, Marcel Dekker, New-York, 1999.
- [2] JIRI MATOUSEK, BERND GÄRTNER : ***Understanding and Using Linear Programming.***, Springer, 2007.
- [3] MIGUEL A. GOBERNA, MARCO A. LÓPEZ : ***Post-Optimal Analysis in Linear Semi-Infinite Optimization***, Springer, Berlin, 2014.
- [4] J. RUCKMANN R REEMSTSEN, J : ***Semi-Infinite programming***, Kluwer Academic Publishers, New-York, 1998.
- [5] CHARLES E. BLAIR : ***A Note on Infinite Systems of Linear Inequalities in $\mathbb{R}^n$*** . Journal of Mathematical Analysis and its Applications 48, 150-154, (1974).
- [6] R. T. ROCKAFELLAR : ***Convex Analysis***, Princeton Mathematical Series, No. 28. Princeton Univ. Press. Princeton, N.J. 1970.
- [7] E. I EKELAND AND T. TEMAM : ***Convex Analysis and Variational Problems***, Studies in Mathematics and its Applications. Vol. 1, North-Holland, American Elsevier new York, 1976.
- [8] ROBERTO LUCCHETTI : ***Convexity and Well-Posed Problems***, Dipto. Matematica Politecnico di Milano, Springer, 2006.
- [9] LUCCHETTI R., PASQUALE, A. : ***A New Approach to a Hyperspace Theory***, Journal of Convex Analysis û April 1996, 04, 173-193(1996)